



Binary Distillation

Pre-Lab Plan

Unit Operations Lab

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Group 1, Thursday, PM section

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1. Experimental Objective

The objective of this lab is to examine a binary distillation column for separating ethanol and water. The phases of this mixture are liquid and vapor, so they are separated by gravity where the liquid goes to the bottom and the vapor rises, this means the two phases travel countercurrent. The McCabe-Thiele method will be used to create graphs which solve countercurrent equilibrium staged separation problems. Two experiments will be conducted. Experiment A: effect of boil-up rate to overall column efficiency, where the boil-up rate of pressure drop over the distillation column will be determined. And Experiment B: distillation at constant reflux ratio, where top and bottom compositions will vary with time by carrying out a distillation at different reflux ratios.

2. Experimental

The apparatus is used for binary distillation processes where the reboiler is filled with a mixture of water and ethanol. Temperature is controlled by power provided by the control panel.

2.1 Apparatus

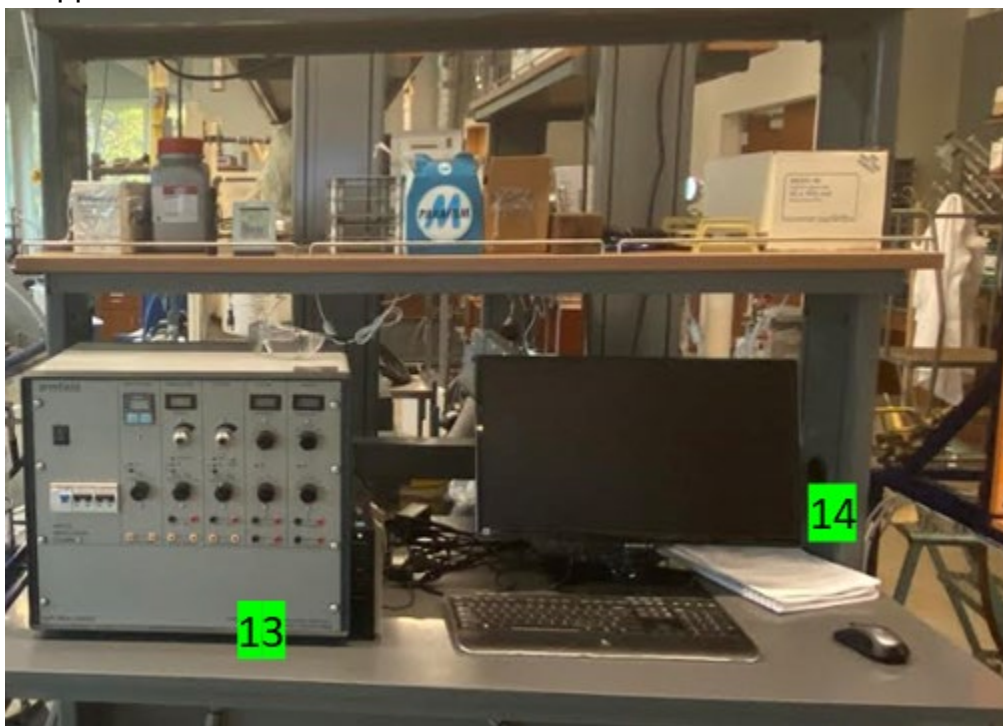


Figure 1: Apparatus for Binary Distillation

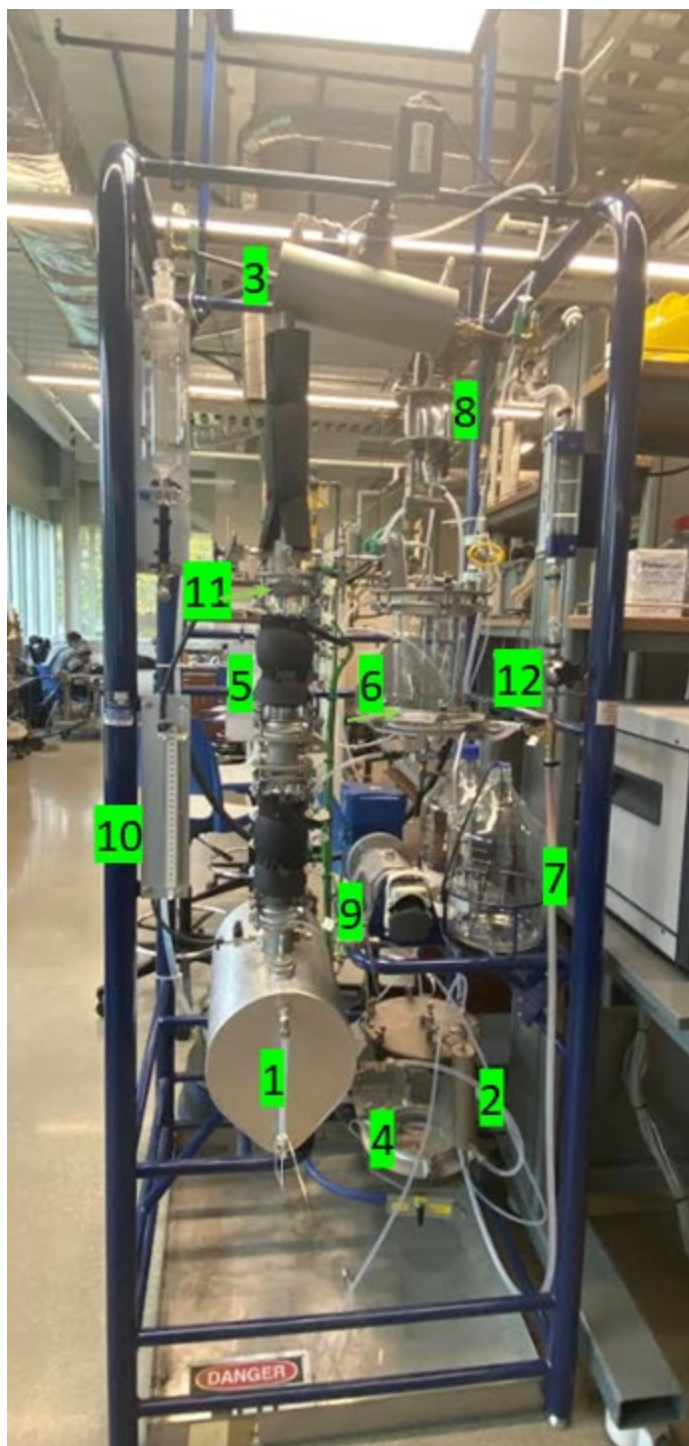


Figure 2: Apparatus for Binary Distillation System continued

Table 1: Apparatus Components

No.	Component Name	Description
1	Reboiler	Heating area for binary mixture
2	Heat Exchanger	Transfer heat
3	Condenser	Compressing area for mixture
4	Bottom Product Receiver	Holds Bottom split solution
5	Column	Distillation column
6	Top Product Receiver	Collection area for split top solution
7	Feed Tanks	Mixture holding area
8	Decanter	Holds Condense liquid
9	Feed Pump	Pump controlled mixture
10	Manometer	Checks pressure
11	Reflux Valve	Separation flow
12	Cooling water Valve	Inlet of cool water
13	Control Panel	Control temperature
14	Computer	Collects temperature profile data

2.2 Materials and Supplies

Table 2: Materials and Supplies

Material / Supplies	Description
Water	Liquid modeled in fluid flow
Electricity	Used to run control panel
Test tubes	Used collect samples
Pocket Refractometer	Measure concentration
Parafilm	to cover test tubes
Graduated Cylinder	Measure water and ethanol
Ethanol	Used for Mixture
Stopwatch	Time Flow rate
Cooling Bath	Cooling test tubes

2.3 Experimental Plan

Experiment A: Effect of boil-up rate and overall column efficiency

- Set-up: Make and load mixture
 - Make 10 L of 10wt% ethanol/water
 - Switch off the reflux ratio timer on the console. Close all valves except V10. Leave V10 open on the reflux pipe
 - Load 10L of mixture through the filler cap on top of the reboiler using a funnel and replace cap firmly after.
- Turn on control panel. Set the temperature switch to T9. Open valve V5 until the flow rate of cooling water (F11) to the condenser is 3 L/min
- On control panel:

- a. Turn power controller for reboiler heating element to 1.5kW and set switch to power on.
 - b. Open V6 and V7 which to manometer then close again
 - c. Note the appearance of Plate 1 and 5 throughout the process
4. When total reflux is reached, turn power controller clockwise until wattmeter reads 0.50kW
5. Measure the boil-up rate at 0.50kW using timed volume collection:
 - a. Adjust V3 so all condensate is diverted into a graduated cylinder at a steady rate pouring the excess into a bottle
 - b. When flow is steady use stopwatch to time how long it takes to collect a certain amount
6. Take readings of pressure drop over the top and the bottom sections by opening V6 and then V7 on the manometer. After reading close the valves in the same order.
7. Analyze samples:
 - a. Keep sample tubes in upright position.
 - b. Measure and record the refractive index for overhead samples
 - c. Measure the composition of the samples using the Refractometer
 - d. Repeat b and c using samples taken from the bottom using V2. IT WILL BE HOT
 - e. Repeat top and bottom sample measurements until 3 samples for top and bottom are taken.
8. Record the temperatures T8 and T1 to calculate the average column temperature
9. Set up boil-up rate in 250W increments up to 1.25kW by adjusting the boiler heater power controller on control panel.
 - a. Record boil-up rate, pressure drops, and top and bottom composition after each increment after it is stabilized

Experiment B: Distillation at Constant Reflux Ratio

1. Using the same ethanol/water mixture already in the reboiler at high temperature from experiment A, set heat controller to 0.75kW to give steady bubbling on all tray and total reflux. Hold for 15 minutes to allow the system to reach an equilibrium condition and record temperature.
2. Set reflux controller to desired value by setting C- and C+, and set reflux valve on using control panel.
3. Measure boil-up rate using timed collection explained in Experiment A 5a. Repeat 3 times and take an average value for boil-up rate
4. Take sample of top using V3. **Never drain condensate return line partially open V3 so liquid is always in line**
 - a. When taking samples drain about 5-10mL before taking next sample in sample glass **But don't drain too much**
5. Sample measurements:
 - a. Keep samples upright
 - b. Record refractive index for the top at the same time as the sample of the bottom through V2. ***It's hot***

- c. Repeat 4 and 5 every 10 mins until five samples of top and five samples of bottom are taken.
- d. Record T8 and T1 to find average temperature of the column.
6. Repeat 2 through 5 for one other different reflux ratio.

Independent & Dependent Variables

Dependent
Pressure
Temperature
Boil-up rate
Concentration
Independent
Reboiler power
Reflux ratio

Data Collection

Table 1: Experiment A

Power (kW)	Boil-up Rate (L/hr)	Pressure drop (cmH ₂ O)			Degree of Foaming on Trays (description)	Top comp	Bottom Comp
		Top	Bottom	Overall			
0.50							
0.75							
1.00							
1.25							

Table 3: Experiment B Data

Reflux ratio	Boil-up Rate (L/hr)	Top comp	Bottom Comp

This table will be filled out during experiment, thus presented in final report.

Project Safety Evaluation

Project Title: Binary Distillation

Date: 10/31/2019

	Yes/No	
Potential electrical hazards?	Yes	Solution spilling on electronics
Potential mechanical hazards?	No	No moving parts so no mechanical hazards
Potential pressure hazards?	No	No, pressure will stay at around 20psi
Potential temperature hazards?	Yes	Collecting samples from the reboiler, sample will be hot
Hazardous/toxic chemicals involved?	Yes	Ethanol is flammable and bad for inhalation
Gloves required?	Yes	Type: Nitrile
MSDS reviewed by all team members?	Yes	MSDS reviewed by all team members
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	100°C	The mixture will reach about 100°C
Pressure (psia)	20 psia	The pressure will stay at around 20psia
Safety Controls	yes/no	If yes, Provide description
Wearing proper PPE, being careful when inserting particles, wear glasses, gloves, face masks if possible	Yes	Be aware of hot temperatures when collecting samples.

I have read relevant background material for the Unit Operations Laboratory entitled: "**Binary Distillation**" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of

my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Alexis Miranda

Signature of team member

10/31/19

Date

Dulce Colindres

Signature of team member

10/31/19

Date

Kathya Chavez

Signature of team member

10/31/19

Date